

[report_combined.pdf](#)

PDF

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Full audit

Worked for 4m 46s >

I audited all 9 pages of the report—equations, numerical formulation, tables, plotted results, claims, internal arithmetic, reproducibility disclosures, and figure presentation. I also cross-checked the methodology against accessible author material from the Liang/Goldsman line of work.

Overall verdict

Conditional pass as an interim qualitative SHE implementation; not yet a defensible full numerical reproduction of Liang et al.

The report is unusually good about disclosing its limitations: reconstructed doping, surrogate band/impact-ionization models, one-way coupling, an unconverged device grid, and an uncalibrated threshold voltage are all stated explicitly. report_combined That honesty materially improves the work.

However, I found **one potentially blocking mathematical issue** in the acoustic energy operator, plus several major numerical and validation gaps. Until those are resolved, I would not describe the Part II result as quantitatively validated or fully reproduced.

Severity	Finding	Audit conclusion
BLOCKER	Acoustic Fokker–Planck operator appears not carrier-number conserving	Eq. (6), as printed, is inconsistent with the report's DOS-weighted particle moment unless the code actually implements a different weighted conservative form
MAJOR	SHE result is not self-consistent	It uses the DD potential without feeding n_{SHE} back to Poisson; this is materially different from the target coupled calculation
MAJOR	Device SHE grid is explicitly unconverged	Peak T_e , G_{ii} , velocity, and hot-tail numbers should not be presented to 3 significant figures
MAJOR	Several critical transport elements are tuned/surrogate	Agreement of magnitudes is not an independent validation when the quantities controlling those magnitudes were adjusted or reconstructed
MAJOR	Surface scattering and impact-ionization implementation are under-specified	The PDF alone is insufficient to reproduce or audit these operators
MAJOR	Degenerate source/drain treated with Boltzmann statistics	$n \sim 10^{20} \text{ cm}^{-3}$ exceeds the report's own N_c by about 3.5×; the contact is outside the nondegenerate regime
MAJOR	Deep-substrate distribution is not actually reproduced	The report calls it carrier-free/noisy, whereas the physical comparison case should give a meaningful near-Maxwellian substrate distribution

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Severity	Finding	Audit conclusion
MAJOR	Validation Table 4 is partly circular	The peak density is largely imposed by the contact Dirichlet value; G_{ii} is compared after using a surrogate/tuned II model
MAJOR	No device current-conservation test	Source/drain current consistency and global carrier balance should be demonstrated before accepting the steady-state solution
MODERATE	“Condition number $\sim 10^{23}$ ” is unsupported	A 10^{23} row-scale disparity is not by itself a condition-number estimate
MODERATE	M-matrix assertion is not established	Positive diagonal coefficients alone do not prove that the assembled matrix is an M-matrix
MODERATE	Pinning orphan states to $f_0 = 0$ can create artificial sinks	This needs a topology/conservation audit or, preferably, removal of the orphan states by construction
MODERATE	Low-density masking affects reported maxima	10^{13} cm^{-3} is a reasonable plotting threshold but needs sensitivity testing
MODERATE	Threshold-voltage mismatch is large	$V_{\text{th}} = 1.64 \text{ V}$ means the nominal 3-V operating point is not electrostatically comparable to a device calibrated near 0.6 V
MINOR	DD-current percentage discrepancy	0.25 versus 0.28 mA/ μm is about 10.7% low , not $\sim 14\%$

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Severity	Finding	Audit conclusion
MINOR	Figure naming/provenance can be tightened	"Report Fig. 4" versus "Liang Fig. 7" is easy to confuse; a commit hash without a retrievable archive is incomplete provenance

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1. The acoustic energy operator is the most serious issue

The report defines the zeroth-order collision term through a **DOS-weighted** balance,

$$\nabla \cdot j_x + \partial_\epsilon j_\epsilon = ZQ[f_0] + \cdots ,$$

and defines electron number by

$$n = \int Z(\epsilon) f_0(\epsilon) d\epsilon.$$

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Yet Eq. (6) is printed as

$$Q_{\text{ac}} = \partial_\epsilon \left[A(\epsilon) \left(\partial_\epsilon + \frac{1}{k_B T} \right) f_0 \right].$$

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Let

$$F = A(\partial_\epsilon + \beta) f_0.$$

The acoustic contribution to particle number is then

$$\int ZQ_{\text{ac}} d\epsilon = \int Z \partial_\epsilon F d\epsilon = [ZF]_{\text{boundaries}} - \int Z' F d\epsilon.$$

Even with zero boundary flux, the second term does **not** generally vanish because Z is energy-dependent.

So, as written, Eq. (6) does not conserve $\int Z f_0 d\epsilon$.

A structurally conservative formulation would instead look like

$$ZQ_{ac} = \partial_{\epsilon} \left[Z A \left(\partial_{\epsilon} + \frac{1}{k_B T} \right) f_0 \right],$$

or equivalently a $1/Z$ -weighted version of that expression, depending on the precise definition of A .

This is not caught by the report's "Maxwellian is an exact null" test. An operator can annihilate one Maxwellian distribution yet still fail to conserve particle number for arbitrary distributions.

This needs to be checked against the code before anything else. If `she2d` already discretizes ZQ_{ac} in conservative flux form and only the paper equation is missing the Z , then this becomes a documentation correction rather than a solver failure. If the code literally implements Eq. (6), it is a blocking physics error.

There is a second concern here: the report says total momentum randomization includes acoustic, impurity, and surface contributions, but then sets $\tau_{ac} = \tau_1$. report_combined If τ_1 is the **total** momentum relaxation time, this makes impurity/surface momentum scattering implicitly influence the reconstructed acoustic energy-relaxation coefficient. The acoustic energy time should be separated from total momentum relaxation, or the approximation and its consequences should be quantified.

2. The mathematical foundation otherwise contains several real checks

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I independently reproduced several numbers in Table 3 from the specifications in the report. The parabolic N_c for $m^* = 0.32m_0$, $g_v = 6$, $T = 300$ K is about $2.726 \times 10^{19} \text{ cm}^{-3}$, consistent with the stated 2.723×10^{19} . The reported Kane velocities at 0.1 and 1 eV are also consistent with the stated dispersion parameters, and the reported n^+ contact potential of about 0.586 V follows from the stated Boltzmann contact formula.

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So these are not decorative “sanity checks”; several of them genuinely reproduce analytically.

The total-energy transformation $H = \epsilon - q\phi$, physical mask, distinction between the reflecting $\epsilon = 0$ boundary and absorbing ϵ_{max} , and integer alignment $E_{\text{op}}/\Delta H = 4$ are also internally coherent.

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One terminology point also checks out externally: accessible Liang-group material explicitly describes solving with the first **four spherical harmonics** and reducing those equations to a second-order self-adjoint equation. Thus the report's $l \leq 1$ “first-order SHE” language is not obviously contradictory; it would simply be clearer to write something such as “ $l \leq 1$, corresponding to four angular coefficients in the 2-D formulation.”

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3. “Reproduction” currently overstates the coupling fidelity

The report explicitly says its Part II SHE calculation is **one-way coupled**: the SHE runs using a drift-diffusion potential, and n_{SHE} is not fed back into Poisson.

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By contrast, accessible Liang-group documentation describes the BTE, Poisson, and hole-continuity equations being solved self-consistently through a Gummel-type process.

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This distinction matters physically. Once the SHE density differs from the DD density, the electrostatic field driving the SHE is no longer the field corresponding to that distribution. The missing feedback can alter pinch-off, channel charge, T_e , and especially the impact-ionization region.

The report is commendably candid about this in Section 7. I would therefore change the headline characterization to “**one-way-coupled qualitative reproduction**” or “**interim SHE emulation**” until the SHE↔Poisson↔hole-continuity loop is closed.

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A useful immediate diagnostic, even before implementing full self-consistency, is to report

$$\frac{\|n_{\text{SHE}} - n_{\text{DD}}\|}{\|n_{\text{DD}}\|}$$

globally and in the active channel. If it is tiny, the one-way approximation has evidence behind it. If it is large, feedback is essential.

4. The source/drain contacts are outside the stated Boltzmann regime

The report uses Boltzmann statistics and source/drain concentrations near 10^{20} cm^{-3} , while its calculated N_c is about $2.86 \times 10^{19} \text{ cm}^{-3}$. report_combined

That gives

$$n/N_c \approx 3.5.$$

This is plainly not the nondegenerate $n \ll N_c$ regime. In fact, the Maxwell-Boltzmann relation gives $(E_F - E_C)/kT = \ln(3.5) \approx 1.25$.

That does not necessarily invalidate a **historically faithful** reproduction if the target model itself made the same approximation. But it does mean the report should stop treating

those contact calculations as strong evidence of physical accuracy. It is a model-consistency benchmark, not a high-doping silicon benchmark.

It is particularly relevant because the SHE contact condition is a Maxwellian normalized directly to n_{contact} . report_combined A Fermi–Dirac/contact-degeneracy sensitivity test would therefore be valuable.

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5. Table 4 overstates the strength of the validation

The reported peaks are $n = 1.05 \times 10^{20}$, $v = 1.23 \times 10^7$ cm/s, $T_e = 3236$ K and $G_{ii} = 1.37 \times 10^{27}$ cm⁻³s⁻¹. report_combined

Their scales are plausible, but the evidentiary value varies sharply.

The peak density is especially weak evidence: the contact boundary itself injects a Maxwellian normalized to the prescribed contact concentration, so observing a global maximum around 10^{20} is largely a boundary-condition consequence. An **interior channel maximum with contact cells excluded**, inversion charge $Q_{\text{inv}}(x)$, or a digitized spatial error against Liang would be much more meaningful.

The G_{ii} agreement is also not independent because the report classifies the impact-ionization rate as a tuned surrogate, while the band and high-energy relaxation physics are also surrogate/reconstructed. report_combined Since impact ionization is exponentially sensitive to the high-energy tail, matching its order of magnitude cannot by itself validate that tail.

The sentence “Every computed magnitude lands on the correct physical scale” is fair as a **sanity statement**. It should not be presented as quantitative validation.

6. The distribution-function reproduction is only partial

Page 8 explicitly says the deep-substrate line is “carrier-free (noisy)” and calls it numerical residual elsewhere. report_combined

That is a meaningful limitation, not merely an unattractive plot. In accessible Liang-group results, the deep-substrate distribution is described as Maxwellian because fields and current density are small there. ResearchGate

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So the report has convincingly demonstrated the **channel hot-tail trend**, but it has not yet reproduced the physically meaningful deep-substrate distribution. I would not count the full distribution figure among the four successfully reproduced paper figures until the substrate panel contains enough actual carrier population to recover the expected equilibrium-like distribution.

Visually, page 8 supports the report's own caveat: the third 3-D surface is dominated by sharp residual-like spikes, unlike the two channel/depth plots.

7. Numerical robustness needs a much stronger section

The engineering log is valuable—it exposes exactly where the solve was difficult.

report_combined But three claims need tightening.

First, the statement that coefficients of order 1 versus 10^{23} imply a “condition number $\sim 10^{23}$ ” is not valid. Matrix scaling disparity and condition number are related but not identical. Say “row-scale disparity of roughly 10^{23} ” unless an actual $\kappa(A)$ or estimator was computed.

Second, “positive-diagonal M-matrix” requires more than a positive diagonal. The sign structure of off-diagonal entries and an appropriate nonsingularity/inverse-positivity or diagonal-dominance argument need to be shown. report_combined If that proof is absent, “monotone positive-diagonal discretization” is safer.

Third, pinning orphan active nodes to $f_0 = 0$ is potentially an **artificial absorbing boundary condition**. If these states should not exist physically, the right solution is to fix the active-mask graph so they are not unknowns. At minimum report how many nodes are pinned, their locations, their integrated carrier weight, and whether removing them changes channel moments.

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8. Convergence is the biggest missing numerical validation after conservation

The report itself says the $40 \times 34 \times 562$ device solve is “smooth but not converged.”

report_combined That makes reporting values such as 3236 K and 1.37×10^{27} misleadingly precise.

The minimum convincing convergence suite would compare at least two finer real-space meshes and two energy spacings, then report changes in drain current, channel charge, peak/centroid T_e , integrated G_{ii} , hot-tail fraction, and peak locations. The tail metrics matter more than visual contour smoothness.

The $\epsilon_{\max}$ check at 300 kV/cm in homogeneous bulk is good, but it should be repeated for the **device impact-ionization integral**. report_combined T_e can be converged while a much rarer > 1 eV tail—and hence G_{ii} —still changes appreciably.

Likewise, the $n > 10^{13} \text{ cm}^{-3}$ mask is sensible for visualization, but the reported maxima should be shown stable under, say, 10^{12} , 10^{13} , and 10^{14} cm^{-3} . Otherwise “physical $T_e \leq 3236$ K” partly reflects the reporting threshold rather than solely the solution.

9. Two transport models are not sufficiently specified to reproduce the work

The taxonomy is one of the strongest parts of the paper because it clearly distinguishes paper-derived, reconstructed, and surrogate elements. report_combined

But two high-impact elements remain too thinly documented.

The text says surface scattering contributes to τ_1 , yet the numerical appendix does not give an explicit surface-scattering law, dependence on distance or normal field, or its parameters. The accessible Liang-group methodology explicitly includes detailed collision physics, so this is a substantive reproducibility gap rather than a cosmetic omission.

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Likewise, the Keldysh impact-ionization surrogate needs its full $1/\tau_{ii}(\epsilon)$ expression, threshold energy, fitted coefficients, units, loss term, and redistribution/gain operator. The present text explains pair bookkeeping but not enough of the operator to independently regenerate G_{ii} .

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10. Current and global conservation checks are conspicuously absent

For a steady-state deterministic transport solver, I would want to see at least three residuals reported: discrete BTE residual, integrated carrier-number balance of each number-conserving collision operator, and terminal current consistency.

The eventual Fig. 8 / I-V sweep is already listed as remaining work. would elevate it from “additional figure” to **core validation**. Source and drain terminal currents should agree to numerical tolerance after accounting for generation/recombination, and integrated impact-ionization generation should reconcile with the hole/substrate-current calculation once full coupling is active.

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That test would also expose the possible Eq. (6) conservation problem immediately.

11. Threshold calibration remains a substantive device mismatch

The report obtains $V_{th} = 1.64$ V but eventually wants calibration to roughly 0.6 V.

At $V_G = 3$ V, the present nominal overdrive is only about 1.36 V.

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Consequently, “3 V / 3 V” is not enough to establish operating-point equivalence with the target device. Charge, lateral field, pinch-off location, hot-electron tail, and impact ionization all depend on the electrostatic calibration.

The remaining-work line “ $V_{th} \rightarrow 0.6 \text{ V}$ ” should also explain that 0.6 V is the **target calibration objective**, because Appendix B says an earlier 0.6-V guess was refuted by a subsequent extraction. report_combined As currently written, those two statements can appear contradictory.

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12. Smaller corrections and presentation issues

Table 3's DD-current comparison should read approximately **11% low**, not ~14%: $(0.28 - 0.25)/0.28 = 10.7\%$. report_combined

The electron “temperature” should preferably be called **mean-energy-equivalent electron temperature**. Its equilibrium value of about 309.7 K rather than exactly 300 K follows from the nonparabolic energy moment being mapped through a $3k_B T_e/2$ convention; without that explanation, readers may interpret 309.7 K as residual numerical heating.

On page 7, the report's “Figure 4” contains plots corresponding to “Fig. 3,” “Fig. 4,” and “Fig. 7” of the target paper. report_combined Page 8 similarly labels report Figure 5 as “Fig. 2.” Explicitly writing “**Liang Fig. 7**” and “**Report Fig. 4**” would eliminate ambiguity.

Finally, the provenance section records a commit hash, package versions, and module names, which is good practice. report_combined But a commit identifier is independently useful only if the repository or permanent archive is available. A publication-ready reproduction package should include the retrievable repository/Zenodo DOI, exact environment/lockfile, commands used for each figure, input files, and ideally raw numerical fields.

What I would require before sign-off

The next revision gate should be: first, verify and, if necessary, correct **DOS-weighted conservation of the acoustic operator** and add explicit number-conservation tests. Then close the SHE–Poisson–hole self-consistency loop, specify the surface and impact-ionization operators completely, recalibrate the device, and conduct real-space/energy-grid plus $\epsilon_{\max}$ convergence studies. After that, validate terminal-current conservation and replace “correct physical scale” comparisons with digitized contour/line overlays and quantitative errors against the target figures.

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Only after those steps would I be comfortable changing the verdict from “**credible qualitative prototype**” to “**numerical reproduction**.”

One limitation of this audit: I could inspect the complete report and independently check its mathematics/numbers, but the actual repository/code was not attached, so the potential Eq. (6) blocker, the assembled-matrix properties, and the claimed code provenance cannot be settled at source-code level from the PDF alone. The accessible Liang-group papers do, however, confirm that self-consistent coupling and the four-harmonic formulation are central features of the original methodology. ResearchGate +1

  ...  Sources

what part of code u need ?

Worked for 5s >

To resolve the **possible blocking issue first**, I only need a small part of the code. Your report says the main relevant modules are `scattering`, `she_bulk`, `bands`, and `she2d`.

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Please send these, in this order:

1. scattering.py — highest priority

- implementation of the acoustic Fokker–Planck operator
- optical phonon in/out operator
- `tau1`
- DOS weighting (z)
- any functions building collision matrices
- impact-ionization rate/operator if it is here

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This will tell me whether the code really implements the printed

$$Q_{ac} = \partial_{\epsilon}[A(\partial_{\epsilon} + \beta)f_0]$$

or whether the **conservative Z -weighted form** is actually implemented and the PDF equation is just missing the weighting. That's the first thing I want to settle. The report identifies this acoustic operator as your reconstruction for numerical irreducibility.

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2. she2d.py — second priority

I want the sections that:

- construct the H -grid and physical mask
- assemble the matrix A
- add spatial diffusion
- add optical/acoustic collision terms
- impose contact and energy-boundary conditions
- detect/pin “orphan” nodes
- do Jacobi row scaling
- solve the linear system
- compute n , velocity, T_e , and G_{ii}

These are the areas behind the conditioning/orphan-node issues described in the report. report_combined

3. `bands.py`

Especially:

- $\gamma(\epsilon)$
- DOS $Z(\epsilon)$
- velocity $v(\epsilon)$
- derivatives of the dispersion
- unit conversions

I need this because whether the collision operator conserves number depends directly on how $Z(\epsilon)$ is defined.

4. `she_bulk.py`

Send the bulk test setup that produces:

- $T_e(0) \approx 309.7 \text{ K}$
- field-heating curves
- Maxwellian null/detailed-balance check
- $\epsilon_{\max}$ convergence

Those are the strongest existing solver tests in the report. report_combined

5. For the full audit afterward: `device.py`, `poisson.py`, `dd.py`, and `diagnostics.py`.

Those let me check the contact potential, reconstructed doping, DD→SHE coupling, current conservation, threshold mismatch, and whether the reported physical comparisons are genuinely independent.

Fastest route: upload `scattering.py`, `she2d.py`, `bands.py`, and `she_bulk.py` first. I can determine the acoustic-conservation question and most of the numerical-assembly questions from just those four.

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If the whole repo at commit `3eb6504` is small, a ZIP of that exact commit is even better; then I can do a source-level audit rather than asking for pieces one at a time.

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    Sources

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bands(1).py
Python

constants.py
Python

dd.py
Python

device.py
Python

diagnostics.py
Python

poisson.py
Python

scattering(1).py
Python

she_bulk(1).py
Python

she2d(1).py

Full audit results

Python

she2d_fig2.py
Python

she2d_figures.py
Python

vth_extract.py
Python

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Worked for 7m 39s >

Yes — this is enough code. I ran a source-level audit and also executed the supplied solver. The picture is much clearer now.

Updated verdict

The acoustic-conservation blocker from my PDF-only audit is resolved. The code implements the conservative Z -weighted form even though Eq. (6) in the report does not show it.

I also reproduced the headline 40×34 device result from your source:

Quantity	My run of your code
Active SHE DOF	328,544
n_{max}	$1.0516 \times 10^{20} \text{ cm}^{-3}$
$T_{e,\text{max}}$	3235.95 K

Quantity	My run of your code
$G_{ii,\max}$	$1.37085 \times 10^{27} \text{ cm}^{-3} \text{ s}^{-1}$
(	v
DD I_D , 40×34	0.2619 mA/μm
Orphan nodes	0

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Those essentially exactly reproduce the numbers in the report. The solver is real and the reported Part II output is reproducible from these files. The code explicitly uses the reported 40×34 grid and 12.5-meV H spacing. she2d

There are, however, some important code-level issues.

Acoustic operator: PASS — report equation needs correction

This was the biggest uncertainty before seeing the code.

Your implementation is

$$-\partial_\epsilon \left[ZA \left(\partial_\epsilon f_0 + \frac{f_0}{kT} \right) \right],$$

not simply the unweighted operator printed in the report. `scattering.py` explicitly constructs `ZA = Z * acoustic_energy_coeff(...)` and applies it as a conservative Scharfetter–Gummel face flux. scattering

I numerically checked the supplied operator. On the 1209-node bulk grid, its discrete conservation error was about 7.6×10^{-17} relative to the largest matrix coefficient, and the Maxwellian-null residual was essentially machine precision under a normalized test.

So the code is right on the issue I flagged; Eq. (6) in the PDF is the problem.

The report should write something equivalent to

$$Q_{ac} = \frac{1}{Z} \partial_{\epsilon} \left[Z A \left(\partial_{\epsilon} + \frac{1}{k_B T} \right) f_0 \right].$$

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That change is important because your report currently defines particle density with the $Z f_0$ measure.

The optical operator also passes the conservation check. It is explicitly constructed as ZQ with paired in/out transitions. scattering I recover the same roughly 2×10^{-5} Maxwellian detailed-balance residual quoted in the report.

Impact ionization: major implementation gap

This is now the biggest physics issue.

Your report describes impact ionization as an actual collision process: remove the energetic primary, create an electron-hole pair, redistribute the resulting electron energy, and have $G_n = G_{p,ii}$. report_combined

The code does not implement that collision operator.

In `she2d.py` , impact ionization appears only after the SHE solution has already been obtained:

```
tii_inv = ...
```

Python

Run

she2d

There is no impact-ionization loss or secondary-electron source in `assemble()` . Likewise, the hole continuity solver contains SRH recombination but no $G_{p,ii}$ term. ^{dd}

So the present calculation is actually:

$$f_0 \longrightarrow \text{post-process } G_{ii},$$

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rather than

$$G_{ii} \leftrightarrow f_0, p, \phi.$$

That means the equal-energy-split model described in the report is **not currently used by the solver at all**.

Consequences: the high-energy tail is not depleted by impact ionization, secondary electrons are not inserted into f_0 , impact-generated holes are not inserted into hole continuity, and their charge does not affect Poisson.

You can still legitimately call the displayed G_{ii} a **diagnostic impact-ionization-rate map**, but not yet a coupled impact-ionization solution.

This also means the statement that Liang Fig. 6 p is simply "available from DD" is problematic: the current DD hole density has no impact-ionization generation source.

The device $\epsilon_{\max}$ boundary is not actually fully absorbing

There is a subtle but real difference between `she_bulk.py` and `she2d.py` .

The bulk solver explicitly imposes

Python

Run

by replacing the last matrix row. she_bulk

That really is an absorbing numerical cutoff.

But in `she2d.py`, when a spatial neighbor is outside the active ϵ region, the code simply skips that face:

```
inactive neighbor -> reflecting no-flux
```

```
she2d
```

The acoustic energy operator likewise skips an inactive adjacent H node. she2d

Only the optical absorption term explicitly permits loss when the upper target lies beyond $\epsilon_{\max}$. she2d

So the 2-D cutoff is currently a **mixed boundary**:

- band edge: reflecting — correct;
- optical transition through $\epsilon_{\max}$: absorbing;
- acoustic flux through $\epsilon_{\max}$: effectively reflecting;
- spatial fixed- H flux through the $\epsilon_{\max}$ surface: effectively reflecting.

That contradicts the `she2d.py` header, which says the upper cutoff is absorbing. she2d

This also means the excellent bulk $\epsilon_{\max} = 3.02/4/5$ convergence result does **not fully validate the device cutoff**, because the two solvers impose the boundary differently.

I would fix this before doing quantitative G_{ii} work.

The velocity figure needs to be recomputed from the discrete face flux

This one produced a measurable difference.

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The solver itself uses finite-volume face fluxes at fixed H . But when you calculate velocity afterward, you reconstruct F_{3d} , put zero in every inactive state, take ordinary central differences, and calculate

$$J = - \int Z D \nabla_H f_0 dH.$$

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she2d

That postprocessing is not exactly the current used by the discrete transport equation. In particular, zeros in inactive states can enter the central differences even where the assembly imposed a reflecting boundary.

I reconstructed the velocity using the **same face-centered $ZD(f_i - f_j)/\Delta x$ fluxes as the actual finite-volume operator**. On your reported 40×34 solution:

- present `moments()` peak: 1.231×10^7 cm/s
- face-flux reconstruction: approximately 9.73×10^6 cm/s

That is large enough that I would not use 1.23×10^7 as a validation number yet.

The best fix is not another gradient formula. Compute Γ_x, Γ_y directly during/from the finite-volume face-flux calculation, integrate those over H , then construct cell-centered velocity. That simultaneously gives you a clean route to **SHE terminal current and current-conservation tests**.

Your Table-I scattering parameters mostly do not drive the solver

This needs clearer wording in the report.

`constants.py` contains the acoustic deformation potential, optical deformation potential, surface-roughness parameter, screening density, surface-acoustic parameter, etc.

constants

But `scattering.py` does not use those mechanisms. The momentum relaxation model is simply

$$\tau_1 = \frac{10^{-14}}{\gamma\gamma'},$$

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and the optical coupling is calibrated through a target 10^{-13} -s scattering time at 0.15 eV.

scattering

So there is currently **no explicit Brooks-Herring impurity contribution, no explicit surface-roughness rate, and no deformation-potential acoustic rate in τ_1 .**

The report's taxonomy partly protects you here because it calls τ_1 a surrogate. But the prose in Section 2.2 says acoustic + ionized impurity + surface terms "enter $1/\tau_1$," which sounds like those terms were actually implemented. They aren't.

I'd change that prose to say the current implementation **replaces their combined momentum relaxation by a calibrated energy-dependent surrogate.**

The Fig. 2 "hot-tail fraction" is not a carrier fraction

The code computes

Python

Run

using raw f_0 .

she2d_fig2

But the number of electrons is weighted by $Z f_0 d\epsilon$, as your solver itself correctly uses for n . she2d

For the exact full solution I ran, at $x \approx 0.6 \mu\text{m}$:

$$\frac{\sum_{\epsilon > 1} f_0}{\sum f_0} = 5.80 \times 10^{-3},$$

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which is your reported number.

The physically meaningful carrier-tail fraction is instead

$$\frac{\int_{\epsilon > 1\text{eV}} Z f_0 d\epsilon}{\int Z f_0 d\epsilon} \approx \mathbf{2.35 \times 10^{-2}}.$$

At the $\sim 0.2\text{-}\mu\text{m}$ line, your raw- f_0 value is 1.12×10^{-8} , whereas the DOS-weighted fraction is about 9.1×10^{-8} .

So I would replace the report's 5.8×10^{-3} number with the DOS-weighted definition, or explicitly call the old quantity an **unweighted f_0 -tail metric**, not a fraction of electrons.

Also, Fig. 2 independently normalizes every depth panel and applies lateral smoothing before plotting. she2d_fig2 That's fine for visualization, but it should be disclosed because it hides the enormous absolute density difference between the channel and substrate.

One reproducibility script is currently stale

`vth_extract.py` really does reproduce $V_{\text{th}} = 1.640 \text{ V}$. vth_extract I ran it and got exactly 1.640 V.

But `diagnostics.py` still defaults to $V_{th} = 0.5$ V for the analytical saturation-current estimate. `diagnostics`

When I execute it as supplied, it gives:

$$I_D/W = 0.9635 \text{ mA}/\mu\text{m},$$

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not the ~ 0.28 mA/ μ m value in the report.

If I change that calculation to the actually extracted $V_{th} = 1.64$ V, it gives

$$0.2851 \text{ mA}/\mu\text{m},$$

which explains the report's ~ 0.28 value.

So this is easy to fix: make `diagnostics.py` use the extracted 1.64 V rather than its stale 0.5-V default.

There's also a sign typo in `vth_extract.py`: it says $dV_{th}/d\Phi_{gate} \approx -1$. `vth_extract`

Since the gate boundary actually uses $V_g - \Phi_{gate}$, `poisson` increasing Φ_{gate} requires increasing V_g by the same amount. The sensitivity should therefore be approximately +1 V/V, not -1 .

Things that passed better than I expected

The band implementation is internally consistent: the Kane γ , its derivative, group velocity and generalized DOS are all constructed coherently, and the code explicitly acknowledges the Kane model as a surrogate for Fiegna/Brunetti. `bands` `bands`

The nonparabolic N_c , parabolic closed-form benchmark, velocity numbers, bulk heating, DD operating point, and the full 328,544-unknown SHE output all reproduce.

The current full matrix also had **zero orphan rows** in my run. So the orphan-node problem appears to be a historical problem that your present formulation/grid has eliminated; the safety pinning code remains but didn't alter this result. she2d

The assembled SHE matrix has positive diagonal and nonpositive off-diagonal entries as intended. I would still avoid stating that this by itself "proves an M-matrix"; the code does not establish the full mathematical M-matrix condition.

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Fix order I'd use

1. **Correct Eq. (6) in the report** to match the already-correct Z -weighted acoustic implementation.
2. **Make the $\epsilon_{\max}$ boundary genuinely absorbing in she2d**, including acoustic and spatial fixed- H fluxes.
3. **Replace velocity postprocessing with finite-volume face-current moments** and add source/drain SHE current conservation.
4. **Implement the actual impact-ionization collision/source operator**, including primary loss + secondary-electron redistribution, then pass $G_{p,ii}$ into hole continuity.
5. Close the **SHE $\rightarrow$ Poisson $\rightarrow$ hole $\rightarrow$ SHE** Gummel loop.
6. Replace the Fig. 2 raw- f_0 tail metric with the **Zf_0 -weighted carrier fraction**.
7. Make the report explicit that current τ_1 is a **single calibrated surrogate**, not a sum of the Table-I impurity/acoustic/surface rates.
8. Fix `diagnostics.py` to use $V_{th} = 1.64$, correct the Φ_{gate} sensitivity sign, and add automated conservation/regression tests.

So the updated assessment is: **the core SHE solve is much more credible than I could establish from the PDF alone, and the acoustic operator is not broken.** The main barrier to calling this a full reproduction is now the missing coupled impact-ionization/self-consistency physics plus the 2-D upper-energy boundary and current-moment implementation.

I don't need any more source files for the core audit. The next useful step would be for me to patch these specific issues in the supplied code, starting with the boundary condition and conservative velocity/current extraction.

    Sources

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please patch

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